

The Gamma-Code Consensus: a dynamical regime model for level of consciousness and global access

Mehmet Oktay Tugan^{1,*}

¹ Independent Researcher, Zurich, Switzerland

ORCID: 0009-0005-8665-3583

* Corresponding author: oktaytugan82@gmail.com

Abstract

The Gamma-Code Consensus (GCC) describes the level of consciousness and global access as a dynamical regime in a network of coupled neural populations. Consciousness is not identified with maximal coherence but with a range of balanced macroscopic integration: global coordination must be present without collapsing into rigid uniformity. The model is formulated as a stochastic network of phase oscillators with three independent levels of control — residual network topology, global gain, and population-specific preservation factors — allowing structural degradation and functional loss of efficacy to be modelled separately. Three operational observables characterize the regime: global coherence, effective dimensionality, and temporal variability. The central contribution is a formal framework that predicts when selective preservation on a residual backbone can produce a transient re-entry into an access-compatible regime. Terminal or paradoxical lucidity is treated as a motivating clinical hypothesis and future empirical target, not as an empirically established result. Numerical simulations on synthetic Kuramoto networks test five model-internal predictions. A retuning-control sweep over four baseline coupling strengths (twenty seeds each), designed as a sweep with the sweep-level statistic as primary analysis, yields a significant main effect of selective preservation over uniform decline (mixed regression, cluster-robust $p = 3.6 \times 10^{-5}$; Cohen's $d = 0.56$). A new stress test on Human Connectome Project-derived Budapest Reference Connectome topologies shows that the selective-backbone re-entry signature can exceed degree-preserving null models and partly survives stricter degree-

plus-strength-preserving controls (best comparison: ΔA_R real-minus-null = 0.091, one-sided Mann–Whitney $p = 0.020$), but the effect is not uniform across all topology and coupling variants. A pilot demonstration on a public EEG dataset (propofol sedation, twenty participants) establishes operational computability and state-discrimination sensitivity (gamma band $d = 0.99$, $p = 6.3 \times 10^{-5}$). Neither the connectome stress test nor the EEG pilot directly validates the selective-preservation mechanism in structurally degraded clinical cohorts, which remains future work.

Contents

Introduction	6
1 Aim and Scope	6
2 Core Intuition	7
3 Core Dynamical Model	8
4 Degradation as Topology and Gain Loss	9
Materials and Methods	9
5 Three Operational Observables	10
5.1 Global Coherence	10
5.2 Effective Dimensionality	10
5.3 Temporal Variability (Metastability)	11
6 Operational Access Region	11
6.1 Multiscale Robustness Formulation	11
6.2 Data-Driven Calibration of Regime Boundaries	12
6.3 Two Positive Reference Levels: $\mathcal{H}_{\text{full}}$ and $\mathcal{H}_+$	12
7 Operational Access Threshold Instead of a Universal K_c	13
8 Basic Mathematical Properties	13
9 Selective-Preservation Re-Entry as a Clinical Hypothesis	13
9.1 Residual Backbone Structure	14
9.2 Selective Gain Preservation Instead of a Global Energy Surge	14
9.3 Three Conditions for a Re-Entry	14
9.4 What the Model Explicitly Does Not Claim	15
10 Predictions	15
11 Empirical Falsifiability	16
Results	16

12 Numerical Validation on Synthetic Networks	17
12.1 Calibration of Regime Boundaries	17
12.2 P2 — Nonlinear Reorganization and Susceptibility Peak	17
12.3 P3 and P4 — Shift of Access Threshold and Bounded Interval (Ensemble)	18
12.4 P5 — Anesthesia Trajectory in Parameter Space (Ensemble)	18
12.5 P6 and P1 — Gain-Mediated Re-Entry on a Residual Backbone (Ensemble) . . .	19
12.6 P6 Retuning Control: Separating Re-Entry from Retuning	21
12.7 HCP-Derived Real-Connectome Stress Test	22
12.8 Assessment of the Numerical Validation	25
13 Pilot Demonstration on Public EEG Data (Chennu 2016)	26
13.1 Dataset and Methodology	26
13.2 Main Finding: Monotonic II-Decline with Sedation Depth	27
13.3 What the Pilot Actually Measures — and What It Does Not	27
13.4 Interpretation in the Context of P1	27
13.5 Robustness of the Pilot Finding	27
13.6 From Ensemble Semantics to Per-Subject Application: A Working Assumption .	28
13.7 Limits and Caveats	28
Discussion	29
14 Relative Originality	29
15 Discussion	29
16 What This Paper Does Not Show	30
17 Conclusion	31
Acknowledgments	32
Data Availability	32
Competing Interests	32
Mathematical Appendix	32

A.1	Assumptions	32
A.2	Elementary Bounds of the Observables	33
A.3	Well-Posedness of the Dynamics	33
A.4	Finite-Horizon Stability	34
A.5	Continuity of the Observables	34
A.6	Residual-Backbone Re-Entry	34
A.7	Corollary: Gain-Mediated Lucidity	35
A.8	What This Appendix Delivers and What It Does Not	35
	Supporting Information	35

Introduction

This introductory block (Sections 1–4) presents the conceptual motivation, the core intuition, and the formal model definition, including the separation of structural degradation from functional loss of efficacy. The observables, access region, predictions, and the proposed re-entry mechanism are developed under Materials and Methods below.

1. Aim and Scope

The GCC is a principled model of level of consciousness and global access. It describes the conditions under which a heterogeneous, noisy, and structurally degradable network of coupled neural populations reaches an integrated but non-rigid macroscopic state. The model does not claim biophysical completeness and does not directly explain the specific content or phenomenal character of individual experiences. Its scope is narrower: it asks when distributed activity passes into a regime in which contents can become globally available. In this sense it is closer to theories of global access, integration/differentiation, dynamic complexity, and disorders-of-consciousness biomarkers than to content or qualia theories in the narrow sense [5, 8, 11, 20, 21, 23, 27, 32].

This restriction is central. It avoids the overreach of turning a simple dynamical model into a full theory of consciousness, and positions the GCC as a testable working model for state transitions such as wakefulness, sedation, anesthesia, severe network degradation, and possible short-term re-integration in structurally compromised systems.

Within this framework, the model formulates a conditional computational prediction:

In degraded neural systems near the lower edge of an access-compatible regime, global access may arise not only from increased overall activity but also from selective preservation of efficacy on a residual backbone structure. Functional re-integration is then predicted to follow from relative gain redistribution onto that backbone rather than from a global energy surge.

This claim is conditional. It concerns degraded systems near the lower edge of the access region, not healthy systems that already lie stably within it. In the healthy range, the relationship between activity and access remains non-monotonic but not inverse. The central empirical consequence of the claim — formulated as testable prediction P1 — remains a future target for

structurally degraded clinical datasets.

Three-layer validation strategy. The paper presents the model in three deliberately separated analysis layers. Section 12 demonstrates *internal consistency* through numerical simulations on synthetic Kuramoto networks: the five directly testable predictions P2–P6 are reproduced under controlled conditions, including a model-internal test of gain-mediated re-entry on a residual backbone. Section 12.7 adds a *topology-generalization stress test* on Human Connectome Project-derived consensus structural connectomes: the same GCC re-entry protocol is run on empirical human connectome topologies and compared with degree-preserving and degree-plus-strength-preserving random controls. Section 13 demonstrates the *operational applicability and state-discrimination sensitivity* of the GCC observables on real biological data: a pilot analysis of the publicly available Chennu et al. 2016 EEG dataset (propofol sedation, $N = 20$) shows that the per-subject calibrated observables discriminate between waking and sedated states. The connectome stress test and EEG pilot are necessary but not sufficient steps toward empirical validation. Direct validation of P1 requires datasets over disorders of consciousness, dementia, or well-documented lucidity-like episodes and remains reserved for future work [14, 18].

2. Core Intuition

The model rests on a simple but strong assumption: consciousness is optimally described neither by maximal desynchronization nor by maximal synchronization. Too little coordination prevents global integration. Too much rigid coordination collapses the dynamics onto a few inflexible modes. The functionally favourable range lies in between: in a regime in which global coordination, reduced but not collapsed dimensionality, and preserved temporal variability co-occur. This intuition aligns with work on communication through coherence, integration versus segregation, metastable brain dynamics, dynamic complexity in conscious states, and entropy-based markers of brain function [7, 10, 12, 13, 16, 31].

The central claim is therefore not “consciousness is synchrony” but:

Level of consciousness and global access arise in a regime of balanced consensus formation within a noisy, heterogeneous network.

3. Core Dynamical Model

We model N neural populations as phase oscillators on a weighted network. The dynamics of population i is given by

$$d\theta_i(t) = \left[\omega_i + K \cdot G(t) \cdot g_i(t) \sum_j W_{ij}(t) \sin(\theta_j(t) - \theta_i(t) - \varphi_{ij}(t)) \right] dt + \sigma_i(t) dB_i(t) \quad (1)$$

where:

- $\theta_i(t)$: phase of population i
- ω_i : intrinsic frequency
- K : global coupling scale
- $G(t) > 0$: global gain or excitability factor
- $g_i(t) \in [0, 1]$: population-specific preservation factor of afferent efficacy
- $W_{ij}(t) \geq 0$: normalized residual topology
- $\varphi_{ij}(t)$: effective phase delays
- $\sigma_i(t) \geq 0$: population-specific noise amplitude
- $B_i(t)$: independent standard Brownian motions

The matrix $W(t)$ is obtained from a raw residual adjacency $A(t)$ by row normalization:

$$W_{ij}(t) = \frac{A_{ij}(t)}{m_i(t)}, \quad m_i(t) = \sum_j A_{ij}(t), \quad (2)$$

provided $m_i(t) > 0$; otherwise $W_{ij}(t) = 0$.

The decisive structural move of the model is the explicit separation of topology W , global excitability G , local preservation g_i , and noise σ_i . This separation is not merely interpretive but mathematically relevant. Row normalization alone makes the incoming structure comparable, but would partially mask structural loss. The factor g_i prevents exactly that: it retains the loss

of afferent efficacy explicitly in the model. The choice of a network-based phase-oscillator framework builds on established results for the synchronization threshold in heterogeneous networks [2, 24].

4. Degradation as Topology and Gain Loss

The GCC does not model degradation as pure edge removal. Structural and functional damage are separated at two levels:

1. $A(t)$ or $W(t)$ describes which connections are topologically preserved.
2. $g_i(t)$ describes how strongly the remaining inputs remain functionally effective.

A simple working definition is

$$g_i(t) = \frac{m_i(t)}{m_i^{\text{ref}}}, \quad (3)$$

where m_i^{ref} denotes the raw incoming mass of the reference connectivity and $m_i(t)$ the currently preserved raw mass. Hence: even if $W(t)$ is renormalized to row-sum one after lesioning, the loss of afferent efficacy does not disappear but is retained in $g_i(t)$.

For degenerative conditions, the model takes the defensive form:

In families of degraded networks, the operational access threshold typically shifts toward higher effective coupling, higher gain, or more favourable noise conditions.

This is a model-internal, empirically testable tendency, not an unconditional universal claim.

Materials and Methods

The following sections (5–11) define the three operational observables, the calibrated access region, the operational access threshold, the basic mathematical properties of the dynamics, the proposed re-entry mechanism, and the predictions derived from the model together with their falsifiability conditions. Additional formal material (proofs of well-posedness, bounds, Lipschitz continuity, stability, and the sufficient condition for re-entry) is provided in the Mathematical

Appendix. Details of the HCP-derived connectome stress test and the EEG dataset are contained in Sections 12.7 and 13.

5. Three Operational Observables

5.1 Global Coherence

$$R(t) = \frac{1}{N} \left| \sum_j e^{i\theta_j(t)} \right| \quad (4)$$

$R(t)$ measures macroscopic phase coherence. High values are consistent with global coordination but are not, on their own, sufficient to diagnose functional global access. Rigid pathological high coherence and flexible integration must be distinguished by further quantities.

5.2 Effective Dimensionality

For a window of length τ_D , define the centred activity

$$X_i(s) = \cos(\theta_i(s)) - \langle \cos(\theta_i) \rangle_{[t-\tau_D, t]} \quad (5)$$

and the unregularized window covariance matrix $\hat{C}_D(t) = \text{Cov}(X_1, \dots, X_N)$. To avoid numerical problems in low-rank states, we use a scaled ridge regularization:

$$\varepsilon_D^\lambda(t) = \max\left(\lambda \frac{\text{tr}(\hat{C}_D(t))}{N}, \varepsilon_{\text{floor}}\right), \quad (6)$$

with a dimensionless parameter $\lambda > 0$ and an absolute floor $\varepsilon_{\text{floor}} > 0$. The regularized covariance matrix is

$$C_D^\lambda(t) = \hat{C}_D(t) + \varepsilon_D^\lambda(t) \cdot I. \quad (7)$$

The effective dimensionality is then the participation ratio:

$$D_{\text{eff}}^\lambda(t) = \frac{[\text{tr} C_D^\lambda(t)]^2}{\text{tr}[(C_D^\lambda(t))^2]} \quad (8)$$

Small values mean compression onto few dominant modes; large values indicate highly distributed activity. The relevant range lies between extreme fragmentation and rigid collapse. Related neural-complexity and signal-diversity measures have been used in consciousness research [19, 26].

5.3 Temporal Variability (Metastability)

For a window τ_M define:

$$M_\tau(t) = \text{Var}(R(s) : s \in [t - \tau_M, t]) \quad (9)$$

M_τ measures the temporal variability of the global coherence level. Small values can indicate either rigid disintegration or rigid high coherence. Only in combination with R and D_{eff} does M_τ become interpretable.

6. Operational Access Region

The GCC does not define an ontological criterion for consciousness but an operational access region in the space of observables. Relative to a pre-specified calibration, window choice, and regularization, a system is internally deemed *access-compatible* if

$$\boxed{R(t) > R_{\min}, \quad D_{\min} < D_{\text{eff}}^\lambda(t) < D_{\max}, \quad M_{\min} < M_\tau(t) < M_{\max}} \quad (10)$$

These inequalities are not a metaphysical definition of consciousness. They are an empirical working description of a dynamical regime that is favourable for global access. The regime statement is thus explicitly data- and scale-relative: it describes not consciousness “as such” but testable conditions of a dynamical range in which global access becomes possible.

6.1 Multiscale Robustness Formulation

Let $\mathcal{T}_D$, $\mathcal{T}_M$, and Λ be finite sets of admissible window lengths and ridge parameters, respectively. For each triple $(\tau_D, \tau_M, \lambda)$ define the indicator

$$\mathcal{A}(t; \tau_D, \tau_M, \lambda) = \begin{cases} 1, & \text{if all three inequalities (10) are satisfied,} \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

The multiscale access index is

$$\Pi(t) = \frac{1}{|\mathcal{T}_D| |\mathcal{T}_M| |\Lambda|} \sum \mathcal{A}(t; \tau_D, \tau_M, \lambda). \quad (12)$$

A state is called window-robustly access-compatible if $\Pi(t) \geq q^*$ for a pre-specified fraction q^* close to 1 (e.g. 0.8).

6.2 Data-Driven Calibration of Regime Boundaries

The regime boundaries should be estimated from a reference ensemble $\mathcal{H}$ of access-compatible state segments. A robust baseline calibration is quantile-based. For a pre-chosen $\alpha \in (0, 0.5)$:

$$\begin{aligned} R_{\min} &= Q_\alpha(R \mid \mathcal{H}), \\ D_{\min} &= Q_\alpha(D_{\text{eff}}^\lambda \mid \mathcal{H}), & D_{\max} &= Q_{1-\alpha}(D_{\text{eff}}^\lambda \mid \mathcal{H}), \\ M_{\min} &= Q_\alpha(M_\tau \mid \mathcal{H}), & M_{\max} &= Q_{1-\alpha}(M_\tau \mid \mathcal{H}). \end{aligned} \quad (13)$$

6.3 Two Positive Reference Levels: $\mathcal{H}_{\text{full}}$ and $\mathcal{H}_+$

For a publication-ready operationalization, the GCC recommends two positive reference levels:

- $\mathcal{H}_{\text{full}}$: healthy or near-healthy access segments with high macroscopic integration.
- $\mathcal{H}_+$: broader positive ensemble including partial access segments (mildly sedated, mildly degraded).

In addition, an exclusion ensemble $\mathcal{U}$ is used. This allows two empirically distinct statements to be separated:

- **full-access-compatible:** $\Pi_{\text{full}}(t) \geq q_{\text{full}}$

- **partial-re-entry-compatible:** $\Pi_+(t) \geq q_+$, even if $\Pi_{\text{full}}(t) < q_{\text{full}}$

For lucidity-related analyses, this separation is crucial.

7. Operational Access Threshold Instead of a Universal K_c

For finite, noisy, and topologically heterogeneous networks, K_c is not a universal single number. The GCC uses an operational access threshold K_c^{op} :

$$K_c^{\text{op}} = \arg \max_K \frac{d\langle R \rangle}{dK} \quad \text{or} \quad K_c^{\text{op}} = \arg \max_K \chi_R(K), \quad \chi_R(K) = \text{Var}_t(R(t; K)). \quad (14)$$

The threshold is thereby treated as an observable regime transition rather than as an a priori constant [2, 24].

8. Basic Mathematical Properties

Under the assumptions $W_{ij} \geq 0$, $\sum_j W_{ij} = 1$, and $|\sin(\cdot)| \leq 1$, the deterministic coupling term is bounded:

$$\left| K \cdot G(t) \cdot g_i(t) \sum_j W_{ij}(t) \sin(\cdots) \right| \leq K \cdot G(t) \cdot g_i(t). \quad (15)$$

For the observables the elementary bounds hold:

$$1 \leq D_{\text{eff}}(t) \leq \text{rank}(C_D(t)) \leq N, \quad (16)$$

and if $R(s)$ is constant on a window $[t - \tau_M, t]$, then $M_\tau(t) = 0$. High coherence alone is therefore not enough; a rigid system is low in metastability.

9. Selective-Preservation Re-Entry as a Clinical Hypothesis

The GCC introduces a gain-based re-entry mechanism that may be relevant to structurally degraded systems. Terminal or paradoxical lucidity is used here as a motivating clinical phe-

nomenon and future empirical target. It is important not to conflate two literatures: the clinical literature on terminal or paradoxical lucidity is thin, heterogeneous, and case-based [3, 15, 22], whereas the literature on high-frequency coherence in the dying brain comes primarily from neurophysiological studies [6, 33]. The present paper does not empirically test terminal lucidity.

9.1 Residual Backbone Structure

Let S be a subset of populations that carries representation-relevant networks. “Residual” does not mean that S is healthy, but that a connected residual scaffold is present on S . An algorithmic definition:

$$S(t; g_*) = \text{largest connected component of the subgraph on } \{i : g_i(t) \geq g_*\}. \quad (17)$$

9.2 Selective Gain Preservation Instead of a Global Energy Surge

The central mechanism is:

A lucidity-like re-entry, if it occurs by the mechanism proposed here, need not correspond to a global energy surge. It may instead reflect short-term functional concentration of remaining efficacy on a residual backbone structure.

The effective afferent drive of population i is proportional to

$$\Lambda_i(t) = K \cdot G(t) \cdot g_i(t). \quad (18)$$

Even if $G(t)$ decreases overall, the difference between preserved and collapsing populations can increase. The result is a temporary reorganization on a residual backbone.

9.3 Three Conditions for a Re-Entry

The model predicts a lucidity-like re-entry only if three conditions are simultaneously satisfied:

1. **Structural residual condition:** A residual backbone structure S is still present.
2. **Gain condition:** For populations in S , $\Lambda_i(t)$ remains sufficiently high during a short interval.

3. **Dynamical admissibility:** Noise, delay dispersion, and frequency spread prevent neither integration nor flexible dynamics.

These conditions are intentionally strict. The model does not claim that *every* dying process produces a final moment of clarity.

9.4 What the Model Explicitly Does Not Claim

The model does *not* claim

- that terminal lucidity has been empirically demonstrated in the present study,
- that every terminal-lucidity episode has the same mechanism,
- that global noise reduction is the sole cause,
- that high coherence alone explains autobiographical memory,
- or that every dying process necessarily produces a final consensus-like peak.

10. Predictions

Six publishable, conservatively stated predictions follow from the model.

P1. Functional re-integration through selective preservation. (*Central empirical target.*) In severely degraded systems, states with reduced overall activity but elevated relative integration within a residual subnetwork should correlate with improved access-compatible dynamics. This prediction is conditional: it concerns degraded systems near the lower edge of the access region, not healthy systems that already lie stably within it.

P2. Critical range rather than arbitrary gradualism. Near the access threshold the system shows a sharp non-linear reorganization. This need not be strictly discontinuous but should not be arbitrarily smooth.

P3. Degradation shifts the access region. In ensembles of degraded networks, K_c^{op} typically shifts toward higher effective coupling or higher gain.

P4. Global access lies in an interval. Neither maximal desynchronization nor maximal rigid synchrony correlates optimally with global access. An intermediate range is expected.

P5. Anesthesia and awakening are trajectories in parameter space. State transitions can be described as motion in the space spanned by W , G , g_i , σ_i , and φ_{ij} .

P6. Lucidity-like re-entry is a future clinical target of P1. If paradoxical or terminal lucidity is driven by the mechanism proposed here, it should occur preferentially when a residual backbone structure and selective gain preservation make a short re-entry possible. It need not be healthy-like; partial re-entry at a lower functional level would also count. P6 is therefore a specific, future empirical target of P1, not a result established by the present dataset.

11. Empirical Falsifiability

The model fails if robust contrary findings emerge:

- Global conscious access systematically occurs without elevated macroscopic integration.
- Severe degradation does not shift the operational access region.
- Rigid high coherence correlates systematically better with global access than an intermediate range does.
- Well-documented lucidity-like episodes in structurally degraded cohorts show no indication of transiently improved global integration.
- The predictions turn out to be highly artifactual without ensemble robustness.

Falsifiability is here not a side issue but part of the model: the GCC does not fail only at a single counter-intuition, but already if its regime boundaries, after calibration and robustness analysis, show no reliable empirical discriminatory power.

Results

The following sections report the numerical validation of predictions P2–P6 on synthetic Kuramoto networks (Section 12), a topology-generalization stress test on HCP-derived consensus

structural connectomes (Section 12.7), and the pilot application of the GCC observables to publicly available EEG data (Section 13).

12. Numerical Validation on Synthetic Networks

To verify the internal consistency of the model and the formal correctness of the predictions in Section 10, we implement the full dynamical system on synthetic networks and test the five model-internal predictions P2–P6 under controlled conditions. P1 remains the central empirical target for structurally degraded biological cohorts; the present simulations test a controlled instantiation of that mechanism, and the EEG pilot tests only operational state sensitivity. This section establishes that (i) the predicted signatures follow from the model equations, (ii) the observables distinguish regimes, and (iii) gain-mediated re-entry is reproducible in simulated data on an ensemble level. This is a *proof of internal consistency*, not an empirical validation against biological data.

12.1 Calibration of Regime Boundaries

We realize the core dynamical model (Section 3) on Watts–Strogatz networks with $N = 100$ nodes, mean degree 10 and rewiring probability 0.1. The coupling matrix W_{ij} is binary. Integration proceeds via Euler–Maruyama with $dt = 0.01$ over $T = 100$ time units. Natural frequencies ω_i are drawn from a standard normal distribution; the gains g_i are initialized to 1 and modified only in specific experiments.

Calibration of the regime boundaries $R_{\min}$, $D_{\min}$, $D_{\max}$, $M_{\min}$, $M_{\max}$ follows the procedure in Section 6. We generate an ensemble of 12 networks under canonical parameters ($K = 2.5$, $\sigma = 0.2$) as the positive reference $\mathcal{H}_{\text{full}}$, extract the observables over a stationary window, and compute the $\alpha = 0.1$ quantiles. The lower bound $M_{\min}$ degenerates to zero in this ensemble configuration, indicating that some healthy systems lock into quasi-stable states. For all subsequent tests, $M_{\min} = 0$ is used; the metastability condition thus effectively reduces to $M_{\tau} < M_{\max}$.

12.2 P2 — Nonlinear Reorganization and Susceptibility Peak

Prediction P2 states that the access indicator Π should increase nonlinearly with the coupling strength K along a sigmoidal curve rather than a linear one. We sweep $K \in [0, 4]$ and fit both a sigmoid and a linear function to the resulting $\Pi(K)$ curve. The sigmoidal fit achieves

Table 1: P3/P4 Ensemble ($N_{\text{seeds}} = 10$ per cell): shift of $\langle K_c^{\text{op}} \rangle$ and maximum access index $\langle \max_K \Pi(K) \rangle$ with increasing degradation. Values are means $\pm$ standard deviations over the seed ensemble. Both quantities behave monotonically under increasing lesion.

Lesion f	$\langle K_c^{\text{op}} \rangle \pm \text{SD}$	$\langle \max_K \Pi(K) \rangle \pm \text{SD}$
0.00	1.38 ± 0.57	0.83 ± 0.29
0.15	1.49 ± 0.44	0.81 ± 0.24
0.30	1.80 ± 0.40	0.71 ± 0.30
0.45	2.15 ± 0.33	0.58 ± 0.38

$R^2 = 0.982$, the linear fit $R^2 = 0.948$; $\Delta R^2 = 0.034$. The curvature is clearly present but not abrupt, consistent with the “nonlinear but not strictly discontinuous” character that the model predicts. A susceptibility peak (standard deviation of R across network realizations) is visible in the midrange of the transition. P2 is confirmed.

12.3 P3 and P4 — Shift of Access Threshold and Bounded Interval (Ensemble)

Predictions P3 and P4 concern the behavior of the access threshold under degradation. We introduce graded degradation by removing a fraction $f \in \{0, 15, 30, 45\}\%$ of network edges (topological degradation). For each (K, f) combination, we run $N_{\text{seeds}} = 10$ independent network seeds and record K_c^{op} (first K -crossing of $R_{\min}$) and $\max_K \Pi(K)$ per seed, then average over the ensemble (Table 1).

P3 is confirmed by the monotonic upward shift of $\langle K_c^{\text{op}} \rangle$: with increasing lesion the system requires stronger coupling to reach the access region. Pairwise Wilcoxon tests between lesion levels show a significant shift for $f = 0.00$ vs. $f = 0.45$ ($p = 0.002$). P4 is confirmed by the bounded maximum of $\Pi(K)$: even at optimal K , no state reaches $\Pi = 1$, and the attainable maximum fraction decreases monotonically from $\langle \max \Pi \rangle = 0.83$ (intact) to 0.58 (45% lesion), a total reduction of -30% (Figure 1). Empirically this means: the access region is a bounded range, not an open half-space, and it quantitatively shrinks under increasing degradation.

12.4 P5 — Anesthesia Trajectory in Parameter Space (Ensemble)

Prediction P5 describes the expected trajectory under a dynamic anesthesia protocol: the state should decrease smoothly during induction, collapse at maximal effect, and recover upon washout. To capture the ensemble-statistical nature of Π correctly, the trajectory was repeated with $N_{\text{seeds}} = 20$ independent network seeds; reported values are means and standard deviations across the seed ensemble. We implement a time-varying gain $G(t)$ with schedule: baseline

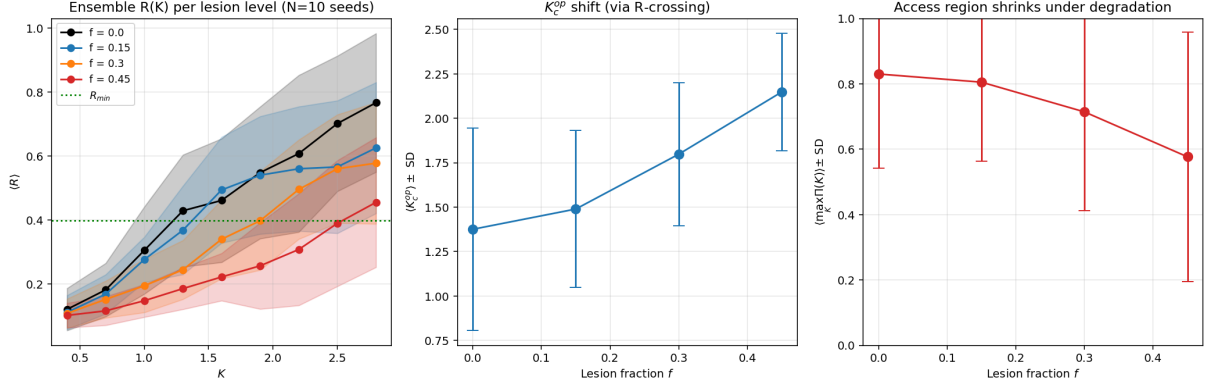


Figure 1: P3 and P4 Ensemble ($N_{\text{seeds}} = 10$): Left — $\langle R \rangle$ vs. K for four degradation levels, shaded bands $\pm$ SD; horizontal line marks $R_{\min}$. Middle — shift of $\langle K_c^{\text{op}} \rangle$ with lesion fraction, error bars SD. Right — maximum access index $\langle \max_K \Pi(K) \rangle$ vs. lesion fraction.

($G = 1.0$, $t < 3s$), induction ($G : 1.0 \rightarrow 0.2$), deep ($G = 0.2$), emergence ($G : 0.2 \rightarrow 1.0$), recovery ($G = 1.0$). The ensemble-averaged access indicator follows the expected course: baseline $\langle \Pi \rangle = 0.47 \pm 0.41$, deep $\langle \Pi \rangle = 0.00 \pm 0.00$, recovery $\langle \Pi \rangle = 0.46 \pm 0.29$. The trajectory in (R, D_{eff}) space leaves the access region and re-enters it smoothly, without jumps (Figure 2). P5 is confirmed: state transitions are motion in parameter space, not jumps between discrete states.

12.5 P6 and P1 — Gain-Mediated Re-Entry on a Residual Backbone (Ensemble)

P6 is the sharpest prediction of the model and tests the central claim P1 in its most specific form: functional re-integration through selective loss, concretely as a transient re-entry on a residual backbone. The system is placed in severe collective degradation, and we test whether a small subnetwork S (30 nodes out of 100), in which the gains are selectively boosted ($g_i : 0.5 \rightarrow 0.8$) while all non- S gains simultaneously collapse ($g_i : 0.5 \rightarrow 0.1$), can transiently reach the access region. The comparison condition is a uniform gain decay in which all g_i drop by the same factor.

To capture the ensemble-statistical nature of Π correctly, both conditions were simulated pairwise with $N_{\text{seeds}} = 20$ independent seeds. Reported values are ensemble means and standard deviations plus a paired statistical test. Results are summarized in Table 2 and Figure 3.

The quantitative difference in the re-entry window $\langle \Pi_S^B \rangle - \langle \Pi_S^A \rangle = 0.122$ is significant by paired t-test ($t = 3.57$, $p = 0.002$) and paired Wilcoxon ($W = 9.0$, $p = 0.006$). The relative increase is +81%, Cohen's $d = 0.82$. Sixteen of 20 seeds show $\Pi_S^B \geq \Pi_S^A$. Crucially, this effect occurs *while*

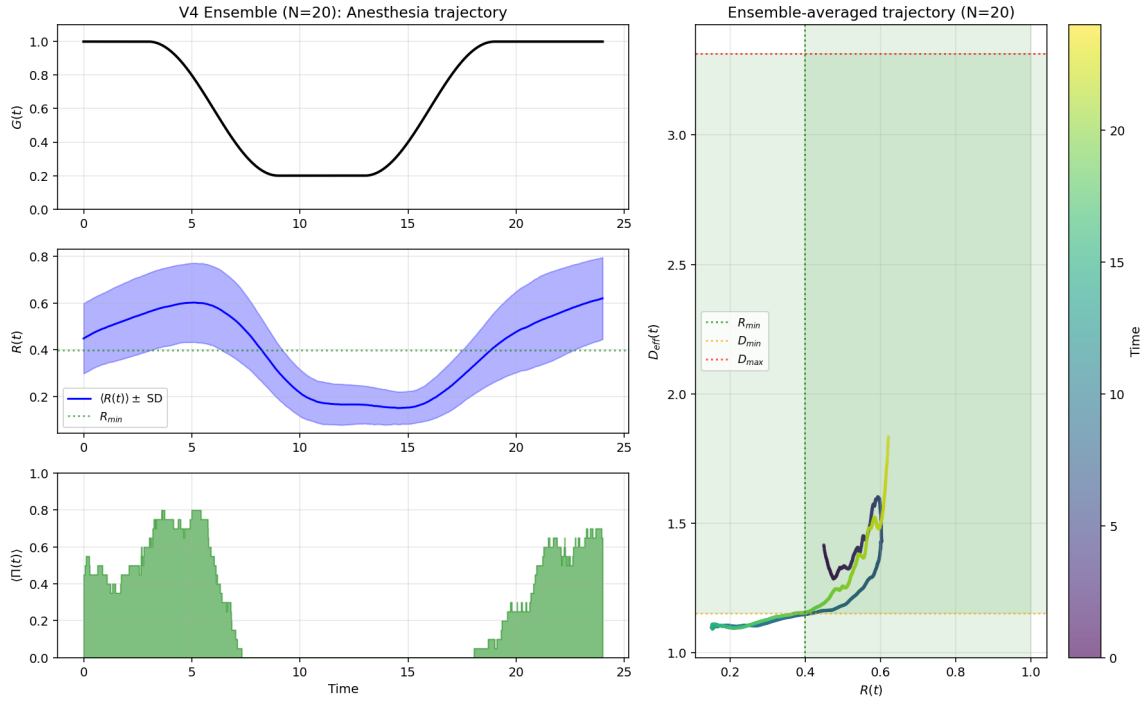


Figure 2: P5: Ensemble-averaged anesthesia trajectory ($N_{\text{seeds}} = 20$). Left top to bottom: $G(t)$ schedule, $\langle R(t) \rangle \pm \text{SD}$, ensemble-averaged $\langle \Pi(t) \rangle$. Right: ensemble-averaged trajectory in (R, D_{eff}) space, color-coded by time; shaded region marks the access region.

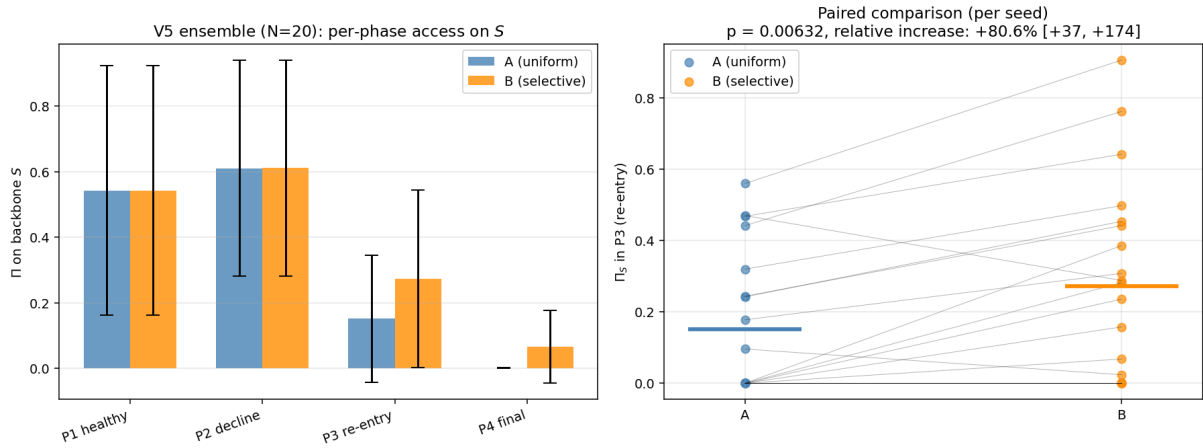


Figure 3: P6/P1: Ensemble results of the gain-mediated re-entry simulation ($N_{\text{seeds}} = 20$). Left: phase-wise Π_S in condition A (uniform, blue) and condition B (selective preservation, orange); error bars show ensemble SD. Right: paired comparison across seeds in phase 3 (re-entry); each line connects a seed pair.

Table 2: P6/P1: Phase-wise access indices Π (fraction of time inside the access region), averaged across $N_{\text{seeds}} = 20$ independent seeds (mean $\pm$ SD). **Critical finding:** In phase 3 (re-entry window), condition B (selective preservation) shows a significantly higher access index on backbone S than condition A (uniform).

Phase	Π A full	Π B full	Π A on S	Π B on S
Phase 1 healthy	0.53 ± 0.38	0.53 ± 0.38	0.54 ± 0.38	0.54 ± 0.38
Phase 2 decline	0.69 ± 0.33	0.69 ± 0.33	0.61 ± 0.33	0.61 ± 0.33
Phase 3 re-entry	0.09 ± 0.12	0.09 ± 0.13	0.15 ± 0.19	0.27 ± 0.27
Phase 4 final	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.07 ± 0.11

overall gain decreases — it arises not from a global energy surge but from selective preservation of efficacy on a subset.

12.6 P6 Retuning Control: Separating Re-Entry from Retuning

A second concern deserves explicit testing. In Table 2, $\langle \Pi_{\text{full}}^A \rangle$ rises from 0.53 (P1 healthy) to 0.69 (P2 decline) in *both* conditions. That is: uniform gain decline alone already improves the access index. Part of the +81% re-entry effect could therefore be explained simply by the system starting at $K = 1.6$ *below* its optimal access range and being shifted into a more favorable regime by uniformly lowering gains (*retuning effect*).

To separate retuning from re-entry, the P6 protocol was repeated at four baseline coupling strengths $K_{\text{baseline}} \in \{1.6, 2.0, 2.5, 3.0\}$ with $N_{\text{seeds}} = 20$ each (Table 3, Figure 4).

Table 3: Retuning vs. re-entry effect across four baseline coupling strengths ($N_{\text{seeds}} = 20$ per K).

K_{baseline}	Retuning $\Delta\Pi$	$\langle \Pi_S^A \rangle$	$\langle \Pi_S^B \rangle$	B–A	Rel. increase	Wilcoxon p
1.6	+0.16	0.15	0.27	+0.12	+81%	0.006
2.0	+0.07	0.26	0.53	+0.27	+102%	0.0009
2.5	+0.19	0.52	0.75	+0.23	+44%	0.004
3.0	+0.15	0.73	0.76	+0.04	+5%	0.54 n.s.

Three descriptive findings follow from this sweep. *First:* the re-entry effect is significant at three of the four K values. *Second:* retuning and re-entry are separate quantities at all K , with the cleanest dissociation at $K = 2.0$ (retuning +0.07, re-entry +0.27, factor 4). *Third:* at $K = 3.0$ the re-entry effect collapses to ceiling level (+5%, n.s.), while retuning persists.

Primary analysis: pooled sweep statistic. To avoid post-hoc selection, we report the entire K -sweep as the main result through a pooled regression with seed clustering. The model $\Pi \sim \text{Condition} \cdot K_{\text{baseline}} + (1|\text{Seed})$ over all 80 paired observations yields a highly significant

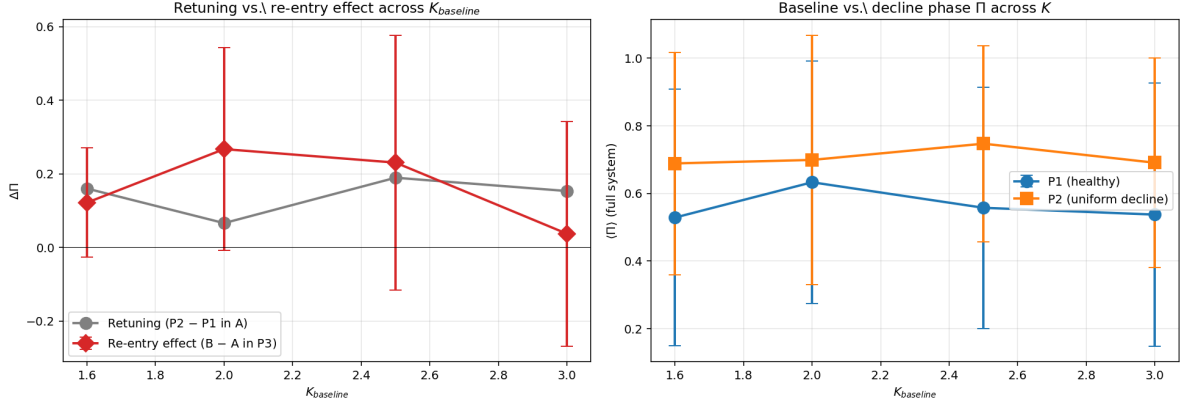


Figure 4: P6 retuning control. Left: retuning effect (gray) and re-entry effect (red) across baseline coupling strength. Right: full-system Π in P1 (healthy) and P2 (uniform decline) across K .

main effect of Condition, $\hat{\beta}_{\text{Cond}} = +0.164$, cluster-robust SE = 0.040, $p = 3.6 \times 10^{-5}$. The Condition $\times$ K interaction is not significant ($p = 0.32$).

Secondary, descriptive plausibility check. As a complementary, non-primary statistic, we report a pooled paired Wilcoxon test across all 80 pairs ($W = 353.5$, $p = 5.5 \times 10^{-7}$, Cohen's $d_{\text{paired}} = 0.56$). We explicitly note that this test pools non-independent observations (each seed appears four times, once per K value) and serves only as a descriptive consistency check.

Interpretation. The re-entry effect is robust across the sweep and not adequately captured by a simple retuning explanation. The result at $K = 2.0$ is retained as a *point of maximum sensitivity* within the sweep, not as a primary result selected from it.

12.7 HCP-Derived Real-Connectome Stress Test

The synthetic validation above uses controlled Watts–Strogatz networks. This is useful for isolating mechanisms, but it leaves open whether the same re-entry signature is specific to the artificial topology. To test topology generalization, we repeated the GCC selective-preservation protocol on Human Connectome Project-derived Budapest Reference Connectome v3.0 consensus graphs [29, 30]. These are not patient data and do not instantiate neurodegeneration. They provide a biologically grounded healthy structural topology on which the model can be stress-tested.

Six HCP-derived consensus variants were analyzed: 20k-fiber tractography at 10%, 25%, and 50% edge-confidence thresholds, 200k- and 1m-fiber 50% variants, and a 20k electrical-connectivity variant. Self-loops in the downloaded edge lists were excluded. Each graph was restricted to its

largest connected component. The residual candidate backbone S was defined a priori as the top 10% of nodes by raw node strength in the empirical graph. The same S was then used in all matched null graphs. This prevents the null controls from being weakened by redefining the backbone after randomization.

Two null ensembles were generated for each connectome variant. The first preserves the binary degree sequence through double-edge swaps and assigns shuffled empirical weights. The second is stricter: after degree-preserving rewiring, empirical weights are iteratively rescaled to approximately match the original node-strength sequence. For each variant, four coupling values ($K \in \{1.6, 2.4, 3.2, 4.0\}$) were tested, with 12 null graphs per null type. The primary topology stress-test statistic was the event-driven change in backbone order advantage,

$$\Delta A_R = (R_S - R_{\bar{S}})_{\text{event}} - (R_S - R_{\bar{S}})_{\text{baseline}}, \quad (19)$$

where R_S denotes the local phase-order parameter on the pre-defined backbone and $R_{\bar{S}}$ denotes the same quantity on its complement.

Table 4: Selected HCP-derived connectome stress-test results against the stricter degree-plus-strength-preserving null model. Positive real-minus-null values indicate a stronger event-driven backbone-order advantage on the empirical topology. Values shown are the five largest real-minus-null contrasts over the pre-specified variant-by- K grid. FC denotes fiber-count weighting.

Connectome variant	K	Real ΔA_R	Null ΔA_R	Real–null, p
1m FC 50%	2.4	0.135	0.043	+0.091, $p = 0.020$
20k FC 25%	3.2	0.109	0.042	+0.067, $p = 0.118$
20k FC 10%	4.0	0.148	0.083	+0.065, $p = 0.097$
200k FC 50%	2.4	0.094	0.031	+0.064, $p = 0.063$
20k FC 50%	2.4	0.128	0.078	+0.050, $p = 0.143$

The result is mixed but informative. Against degree-preserving controls, the empirical connectomes showed a larger ΔA_R in 21 of 24 variant-by-coupling comparisons, with 8 of 24 one-sided Mann–Whitney tests below $p < 0.05$. Against the stricter degree-plus-strength-preserving controls, the empirical connectomes remained positive in 17 of 24 comparisons, but only 1 of 24 reached $p < 0.05$ and 4 of 24 reached $p < 0.10$. The strongest effect occurred for the 1m-fiber 50% consensus graph at $K = 2.4$ (real $\Delta A_R = 0.135$, null $\Delta A_R = 0.043$, real-minus-null = 0.091, $p = 0.020$).

The structural sufficient-condition diagnostic was less supportive. The appendix defines the external residual mass $\alpha_S = \max_{i \in S} \sum_{j \notin S} W_{ij}$ as a conservative sufficient-condition term: smaller

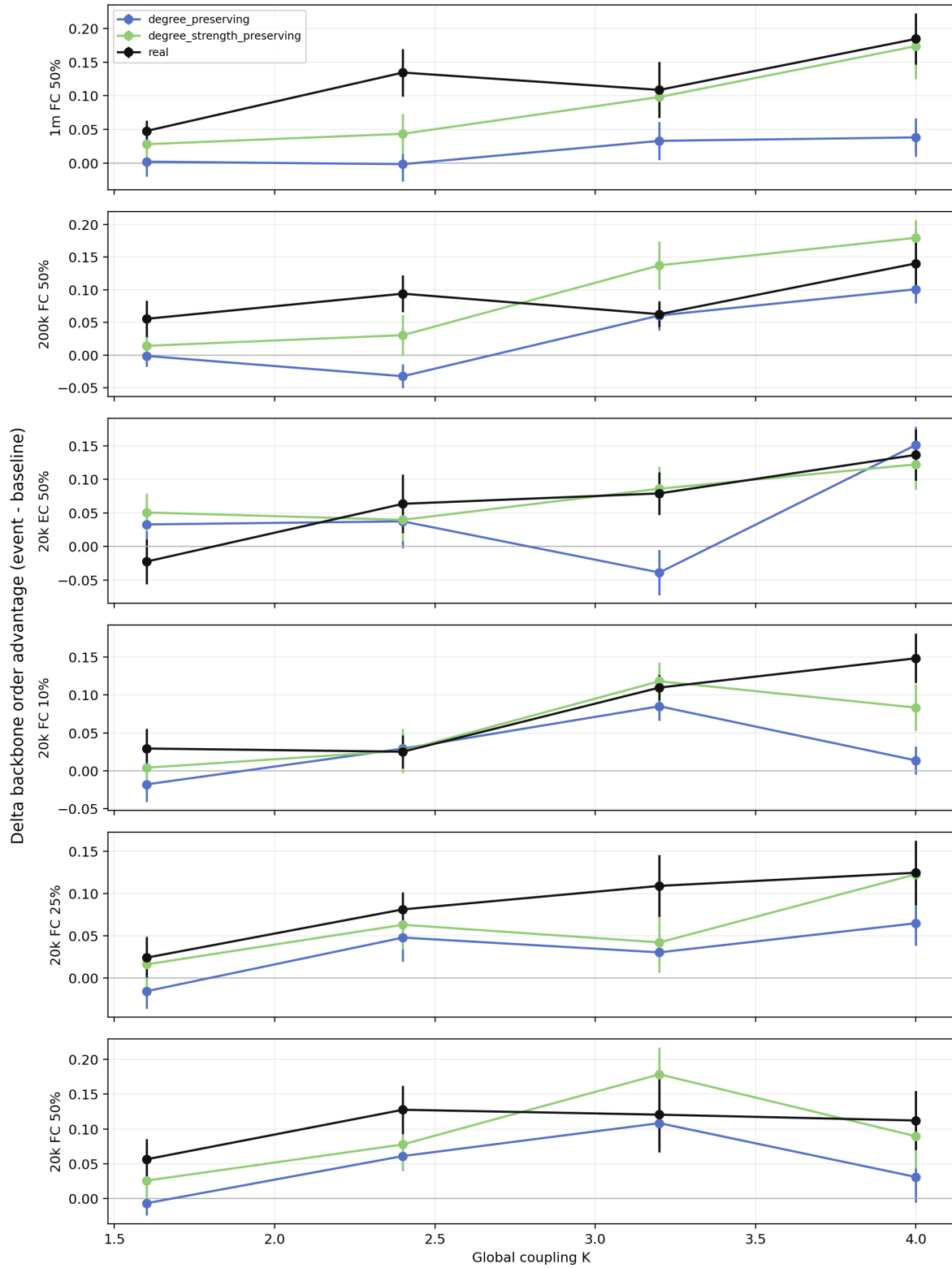


Figure 5: HCP-derived connectome stress test. Each panel shows the event-minus-baseline change in backbone order advantage ΔA_R across coupling values for the empirical connectome, degree-preserving null graphs, and degree-plus-strength-preserving null graphs. Error bars show standard errors across stochastic replicates or null realizations.

values mean weaker perturbation from outside the residual backbone. In the HCP-derived stress test, α_S was generally *not* smaller in empirical graphs than in degree-plus-strength-preserving nulls. For example, in the strongest dynamic comparison (1m FC 50%), α_S was 0.974 for the empirical graph versus a null mean of 0.861. Thus, the observed dynamic advantage cannot be claimed to arise from a cleaner structural isolation of the backbone under this simple top-strength definition.

This analysis therefore strengthens the paper in a specific and limited way. It shows that the GCC selective-preservation signature is not confined to toy synthetic graphs and can appear on HCP-derived human structural topologies. However, because survival against degree-plus-strength-preserving controls is partial rather than uniform, and because the conservative α_S condition is not improved in the empirical graphs, the result should be read as a topology-generalization stress test rather than as biological validation of the P1 mechanism.

12.8 Assessment of the Numerical Validation

Five of the six predictions are directly testable within the synthetic network model. P1 receives model-internal support through the P6 re-entry simulation, but it remains empirically untested in the present biological data.

- **P2:** Nonlinear reorganization, not strictly discontinuous.
- **P3:** K_c^{op} shifts monotonically from 1.38 (intact) to 2.15 (45% lesion), using the ensemble values reported in Table 1.
- **P4:** $\langle \max_K \Pi(K) \rangle$ stays below 1 and decreases monotonically with degradation.
- **P5:** Ensemble-averaged trajectory through the access region runs smoothly.
- **P6:** Differential degradation produces an ensemble-level transient re-entry within the model. Mixed regression over the sweep: $\hat{\beta}_{\text{Cond}} = +0.164$, $p = 3.6 \times 10^{-5}$, $d_{\text{paired}} = 0.56$.

Limits of this validation. (i) The main mechanism tests remain simulations, not direct clinical validation. The HCP-derived stress test uses biologically grounded healthy structural topologies, but not patient data, neurodegeneration, or lucidity episodes. (ii) Calibration and tests use the same ensemble paradigm. (iii) The precision of the regime boundaries depends

on α and window length. (iv) The synthetic re-entry was tested with a fixed backbone size of 30 nodes and a specific boost factor; full parameter sensitivity remains an explicit item of the next research phase. (v) The lower metastability bound degenerates to zero, reducing the formally three-dimensional access criterion to an effectively two-dimensional one. (vi) The real-connectome stress test produced mixed results against the stricter degree-plus-strength-preserving nulls and did not support a simple structural-isolation interpretation via lower α_S .

13. Pilot Demonstration on Public EEG Data (Chennu 2016)

The analyses in Sections 12 and 12.7 operate on simulated dynamics, first on synthetic graphs and then on HCP-derived consensus structural topologies. We now carry out a pilot demonstration on real biological EEG data — a *proof of concept of the analysis pipeline*, not a definitive empirical validation of the model. The pilot fits within the broader literature on anesthesia and consciousness-state research [1, 4, 25].

13.1 Dataset and Methodology

Ethics statement. This study performs secondary analysis of the publicly available EEG dataset released by Chennu et al. [9] under a CC BY 2.0 UK license (University of Cambridge Data Repository, DOI: 10.17863/CAM.68959). The original data collection was approved by the relevant local research ethics committee in Cambridge, UK, as reported in Chennu et al. [9], and was performed in accordance with the Declaration of Helsinki; written informed consent was obtained from all participants by the original investigators. No new data were collected for the present work, and the secondary analysis reported here falls within the scope of the original ethics approval and the data license terms. No additional ethics approval was sought for this re-analysis.

We use the public EEG dataset from Chennu et al. [9]: 20 healthy participants, 91-channel EEG at 250 Hz sampling rate, four resting states per person. The processing pipeline comprises bandpass filtering 1–45 Hz, notch filter at 50 Hz, average re-referencing, and resampling to 250 Hz. We extract phases in two frequency bands separately: *gamma* (30–45 Hz) as the theory-driven choice and *alpha* (8–15 Hz) as the empirically dominant band in Chennu’s original finding. Analytic phase is obtained via the Hilbert transform [17, 28].

From the phase matrices, the GCC observables $R(t)$, $D_{\text{eff}}(t)$, and $M_\tau(t)$ are computed with

$\tau_D = 200$ ms and $\tau_M = 500$ ms. Regime boundaries are calibrated by a quantile procedure ($\alpha = 0.1$) from *each participant's individual baseline recording*.

13.2 Main Finding: Monotonic Π -Decline with Sedation Depth

Figure 7 shows the distribution of Π per sedation level, for both bands. Table 5 summarizes group means and paired tests.

Table 5: Access indicator Π per sedation level and paired Wilcoxon signed-rank tests (baseline vs. moderate). $N = 20$ participants.

Band	Π_{base}	Π_{mild}	Π_{mod}	Π_{rec}	$\Delta\Pi$ (base–mod)	p -value
Gamma	0.661	0.604	0.543	0.569	+0.118	6.3×10^{-5}
Alpha	0.659	0.641	0.561	0.632	+0.098	0.0037

The main effect is consistent and highly significant in both bands. In the gamma band, baseline-relative Π decreases from 0.661 to 0.543 under moderate sedation (Cohen's $d = 0.99$). 18 of 20 participants show the decline. In the alpha band the decline is 0.098, $d = 0.69$.

13.3 What the Pilot Actually Measures — and What It Does Not

The access index Π here is a *baseline-relative state-deviation metric*, not an *absolute* measure of global access. Baseline Π is by construction close to 0.58–0.66 for every participant — a mathematical artifact of the calibration. The pilot does *not* show that the brain has a specific absolute Π value in waking. It shows that the sedated state exits the subject-specifically calibrated waking window with high effect size: a genuine and interpretable finding, but limited in scope.

13.4 Interpretation in the Context of P1

The pilot does not directly test P1. Propofol sedation modulates effective coupling in healthy subjects; it does not instantiate structural degradation, residual topology loss, or selective preservation on a damaged backbone. Recovery data show that the measured effect is at least partly reversible, compatible with the conditional nature of the GCC regime but not validation of the re-entry mechanism.

13.5 Robustness of the Pilot Finding

The main effect was subjected to bootstrap, leave-one-out, and permutation tests (10,000 re-samples each). All three robustness checks confirm the effect: bootstrap 95 % CI excludes zero

in both bands; leave-one-out maximum p -values 1.26×10^{-4} (gamma) and 7.14×10^{-3} (alpha); permutation $p < 10^{-4}$ (gamma), $p = 0.0042$ (alpha).

A full hyperparameter sweep ($\alpha \in \{0.05, 0.10, 0.15, 0.20\}$, $\tau_D \in \{0.1, 0.2, 0.4\}$ s, $\tau_M \in \{0.3, 0.5, 1.0\}$ s; 36 combinations per band) shows that all 36 of 36 combinations reach $p < 10^{-3}$ in gamma and all 36 reach $p < 0.05$ in alpha. The sign of the effect is preserved in 100% of configurations.

Held-out baseline-window calibration is not reported. The available frozen aggregate result files do not contain window-resolved baseline observables. A stronger follow-up pipeline should split each baseline recording into calibration and held-out test windows. This missing analysis is explicitly treated as a limitation.

13.6 From Ensemble Semantics to Per-Subject Application: A Working Assumption

The simulation Π is an expectation-value-stable quantity, while the EEG Π is calibrated per participant from a single recording. The bridge rests on a *working assumption*: a longer EEG recording under constant conditions acts as effective ensemble averaging. This is an ergodicity-type stipulation, neither derivable from the model nor evidenced by our data here. A reviewer who criticizes this step is correct: the bridge is conceptually fragile and rests on a testable, but in this paper untested, assumption.

13.7 Limits and Caveats

1. **In-sample calibration.** Baseline Π near 0.66 is a mathematical artifact, not a measured result.
2. **No held-out baseline-window validation.** The current pilot uses the same baseline recording to calibrate.
3. **Channel level, no source reconstruction.** Volume conduction is not removed.
4. **Hilbert phases as oscillator-phase proxy.** Theoretically incomplete approximation.
5. **Replication on independent datasets pending.**
6. **Pilot does not test the central P1 mechanism directly.** Healthy subjects under

propofol sedation is state modulation, not structural degradation.

7. Ergodicity-type working assumption untested.

The pilot demonstrates the applicability of the observables and their state-discrimination sensitivity — necessary but not sufficient preconditions for a later direct P1 validation.

Discussion

The following sections discuss the relative originality of the GCC framework, the main interpretation of the findings, and a concluding summary with explicit boundaries.

14. Relative Originality

The GCC claims no absolute monopoly on its components. Integration, synchrony, critical dynamics, global availability, metastability, and complexity-based markers are each well established in the literature [7, 8, 10–13, 16, 19, 26, 32]. The relative distinctive feature lies in their explicit assembly into a single operational regime model for global access under degradation, extended by a sweep-controlled test of selective-preservation re-entry and a topology-generalization stress test on HCP-derived structural connectomes.

The revised scope is deliberately mesoscopic and computational. The paper does not attempt to adjudicate between substrate-level theories of consciousness or to solve the hard problem. It asks a narrower question: whether access-compatible dynamics can be operationalized through global coherence, effective dimensionality, and temporal variability, and whether selective preservation on a residual backbone can be distinguished from generic retuning in controlled simulations and then stress-tested on empirical human connectome topology.

15. Discussion

The central computational claim is that functional re-integration in degraded neural systems can arise from selective preservation and gain redistribution, not only from increased global activity. Embedded in the broader framework of balanced consensus formation:

Level of consciousness and global access emerge in a critical dynamical regime in which residual network topology, effective gain structure, noise, and delays jointly

produce sufficient but non-rigid macroscopic integration.

The strongest immediate contribution is methodological. The paper provides an open computational framework, explicit operational observables, a retuning-control design, a real-connectome topology stress test, and a reproducible EEG pilot pipeline. Model-internal simulations show that the proposed dynamics generate threshold shifts, bounded access intervals, anesthesia-like trajectories, and transient residual-backbone re-entry. The HCP-derived connectome analysis shows partial topology generalization: the re-entry signature often exceeds degree-preserving nulls and partly survives degree-plus-strength-preserving nulls, but the result is not uniform and does not support a simple structural-isolation explanation. The EEG pilot shows that the observables are computable on real data and detect propofol-induced state changes under subject-specific calibration.

The biological interpretation must remain narrower. The Chennu dataset contains healthy participants under propofol sedation. The Budapest Reference Connectome provides healthy structural topology, not diseased or degenerating brain data. Neither dataset contains structural degradation, dementia, disorders of consciousness, or lucidity episodes. For that reason, these empirical components cannot validate the central selective-preservation mechanism. A stronger empirical test would require structurally degraded cohorts [14, 18, 21], pre-registered access-region calibration, source-level or leakage-corrected connectivity, and independent held-out validation.

16. What This Paper Does Not Show

To make the evidential boundaries explicit, the present paper does *not* show:

1. that terminal or paradoxical lucidity is caused by, or even necessarily involves, selective preservation on a residual backbone — the present work treats this only as a future empirical hypothesis;
2. that the EEG pilot tests structural degradation, residual topology, terminal lucidity, or the P1 mechanism directly;
3. that the HCP-derived connectome stress test tests structural degeneration, lucidity, or clinical re-entry directly;

4. that the synthetic simulations establish biological truth rather than model-internal consistency;
5. that the HCP-derived topology result is uniformly positive against degree-plus-strength-preserving null models;
6. that subject-specific EEG calibration defines an absolute cross-subject scale of global access;
7. that channel-level Hilbert phases are identical to the theoretical oscillator phases of the model;
8. that the proposed access index is already a validated biomarker.

These limits are not incidental. They define the next empirical step: a preregistered multi-dataset study in structurally degraded clinical cohorts, with independent calibration, source-level or leakage-corrected measures of functional integration, and structural or functional backbone definitions that are specified before outcome testing.

17. Conclusion

The GCC is not a full model of consciousness and does not empirically explain terminal lucidity in the present study. It is a formal, operationalized, and falsifiable model of access-compatible dynamics. Its principal theoretical move is the separation of residual topology, global gain, and local efficacy preservation. Its strongest computational result is that selective preservation can produce residual-backbone re-entry beyond generic retuning in a controlled simulation sweep. The new HCP-derived connectome analysis adds a useful but bounded stress test: the signature can appear on empirical human structural topology, yet the evidence is mixed under stringent null controls. Its strongest biological-data result is more limited: the proposed observables are computable on EEG data and sensitive to propofol-induced state transitions. Direct validation of the selective-preservation mechanism requires structurally degraded clinical datasets and remains future work.

Acknowledgments

The author thanks the Chennu et al. (2016) team for making their EEG dataset publicly available, which made the pilot demonstration in this study possible. The author also acknowledges the Budapest Reference Connectome project for providing public HCP-derived consensus structural connectomes.

Data Availability

The EEG dataset re-analyzed in the pilot part of this study is publicly available from the University of Cambridge Data Repository (9, DOI: 10.17863/CAM.68959) under a CC BY 2.0 UK license. The HCP-derived connectome stress test uses public Budapest Reference Connectome v3.0 graph downloads [29, 30]; the exact download URLs, graph parameters, preprocessing steps, and generated matrix metadata are recorded in the accompanying data manifest. All author-generated GCC simulation code, the per-subject EEG analysis pipeline, the HCP-derived connectome preparation and stress-test scripts, the retuning-control sweep scripts, the statistical analysis code, derived numerical result tables, and figure-generating scripts are publicly available at https://github.com/oktaytugan82/gcc_simulation and archived on Zenodo under the GCC software concept DOI 10.5281/zenodo.19744969. The release corresponding to the present manuscript is v1.1.0.

Competing Interests

The author has declared that no competing interests exist.

Mathematical Appendix

This appendix supplements the GCC with mathematical results that address three common lines of attack: (1) Is the dynamics well-posed? (2) Are the observables controlled? (3) Can the gain-based re-entry idea be formulated as a sufficient condition?

A.1 Assumptions

We consider, over a finite horizon $t \in [0, T]$, the GCC dynamics (1) under the following assumptions.

A1. Structure. $W_{ij}(t) \geq 0$, $\sum_j W_{ij}(t) = 1$ for all i with non-empty rows, $W_{ii}(t) = 0$.

A2. Boundedness. There exist finite constants $G_{\max}$, $\Sigma_{\max}$, $\Phi_{\max}$ such that

$$0 \leq G(t) \leq G_{\max}, \quad 0 \leq g_i(t) \leq 1, \quad 0 \leq \sigma_i(t) \leq \Sigma_{\max}, \quad |\varphi_{ij}(t)| \leq \Phi_{\max}.$$

A3. Temporal regularity. All control functions are measurable and bounded; for stability statements, piecewise continuity suffices.

A4. Observation windows. D_{eff}^λ and M_τ are computed on fixed window lengths $\tau_D > 0$ and $\tau_M > 0$, with ridge regularization according to (6).

A.2 Elementary Bounds of the Observables

Proposition A.1 (Elementary bounds). *For all times t :*

$$0 \leq R(t) \leq 1, \quad 1 \leq D_{\text{eff}}(t) \leq N, \quad 0 \leq M_\tau(t) \leq \frac{1}{4}. \quad (20)$$

Proof sketch. $R(t)$ is the modulus of a mean of unit complex numbers. $D_{\text{eff}}(t)$ is the participation ratio of a positive semi-definite matrix. $M_\tau(t)$ is the variance of a time series with values in $[0, 1]$; its maximum is $\frac{1}{4}$. $\square$

A.3 Well-Posedness of the Dynamics

Theorem A.2 (Global Lipschitz property and well-posedness). *Define the drift vector $b(\theta, t)$ component-wise by*

$$b_i(\theta, t) = \omega_i + K G(t) g_i(t) \sum_j W_{ij}(t) \sin(\theta_j - \theta_i - \varphi_{ij}(t)). \quad (21)$$

Then, for all $\theta, \psi \in \mathbb{R}^N$ and all $t \in [0, T]$,

$$\|b(\theta, t) - b(\psi, t)\|_\infty \leq 2|K| G_{\max} \|\theta - \psi\|_\infty. \quad (22)$$

Under A1–A3, the GCC dynamics has a unique global strong solution on $[0, T]$ for every initial condition $\theta(0)$.

Proof sketch. Using $|\sin(x) - \sin(y)| \leq |x - y|$ and $\sum_j W_{ij}(t) = 1$:

$$|b_i(\theta, t) - b_i(\psi, t)| \leq |K| G_{\max} \cdot 2 \|\theta - \psi\|_{\infty}. \quad (23)$$

Standard SDE results yield existence and uniqueness. $\square$

A.4 Finite-Horizon Stability

Theorem A.3 (Stability under drift perturbations). *Consider two systems driven by the same Brownian motion with globally L -Lipschitz drifts b and $\tilde{b}$ satisfying $\|b(x, t) - \tilde{b}(x, t)\|_{\infty} \leq \varepsilon(t)$. Then*

$$\|\theta(t) - \vartheta(t)\|_{\infty} \leq e^{Lt} \left(\|\theta(0) - \vartheta(0)\|_{\infty} + \int_0^t \varepsilon(s) \, ds \right) \quad (24)$$

Proof sketch. The diffusion terms cancel in the difference equation. Gronwall's inequality applied to the deterministic remainder yields the claim. $\square$

A.5 Continuity of the Observables

Proposition A.4 (Local Lipschitz property). *For fixed τ_D , τ_M , λ , and $\varepsilon_{\text{floor}}$, the observables $R(t)$, $D_{\text{eff}}^{\lambda}(t)$, and $M_{\tau}(t)$ are continuous functions of the window trajectory. On compact tubes there exists a constant L_{obs} such that*

$$\|O(\text{traj}_1) - O(\text{traj}_2)\|_{\infty} \leq L_{\text{obs}} \cdot \|\text{traj}_1 - \text{traj}_2\|_{\infty}. \quad (25)$$

A.6 Residual-Backbone Re-Entry

Theorem A.5 (Sufficient condition for re-entry). *Let $S = S(t_1; g_*)$ be the residual backbone structure as in (17). On $I = [t_1, t_2]$, consider the system restricted to S with external coupling switched off. Define the external residual mass*

$$\alpha_S(t) = \max_{i \in S} \sum_{j \notin S} W_{ij}(t). \quad (26)$$

Assume: (i) the reduced system lies within the access region with safety margin ρ ; (ii) initial states coincide; (iii) L_{obs} exists; (iv)

$$\int_{t_1}^{t_2} \alpha_S(s) \, ds < \frac{\rho}{L_{\text{obs}} \cdot |K| \cdot G_{\text{max}} \cdot e^{2|K|G_{\text{max}}(t_2-t_1)}}. \quad (27)$$

Then the full system restricted to S remains in the access region throughout I .

Proof sketch. Apply Theorem A.3 with $L = 2|K|G_{\text{max}}$ and $\varepsilon(t) = |K|G_{\text{max}}\alpha_S(t)$, then use L_{obs} . $\square$

A.7 Corollary: Gain-Mediated Lucidity

Corollary A.6. *If a subset S and an interval I exist on which $\alpha_S(t)$ is small, the effective drives $\Lambda_i(t)$ for $i \in S$ stay above the reduced access threshold, and the reduced dynamics on S remains in the access region with margin, then a transient re-entry of the full system onto S is internally sufficiently possible within the model.*

A.8 What This Appendix Delivers and What It Does Not

Delivers. Well-posedness of the GCC as an SDE; controlled bounds for the core observables; finite-horizon stability; a sufficient re-entry condition.

Does not deliver. No mathematical proof that real terminal lucidity exists; no general theorem that every form of dementia shifts the access region; no proof of a universal K_c ; no identifiability result for real EEG/MEG data.

Supporting Information

S1 Code. **GCC simulation and analysis code** (`gcc_simulation_v1.1.0.zip`). Complete Python codebase for the GCC dynamical model, per-subject EEG analysis pipeline, ensemble simulations, V6 retuning-control sweep, HCP-derived Budapest Reference Connectome preparation, degree- and degree-plus-strength-preserving null-model stress tests, mixed-effects/statistical analysis, and figure-generation scripts. Includes `requirements.txt`, data manifests, derived result tables, and `README.md` with reproduction instructions. Corresponds to release v1.1.0 of the public repository https://github.com/oktaytugan82/gcc_simulation,

archived on Zenodo under the GCC software concept DOI 10.5281/zenodo.19744969.

References

- [1] Alkire, M. T., Hudetz, A. G., & Tononi, G. (2008). Consciousness and anesthesia. *Science*, **322**(5903), 876–880. <https://doi.org/10.1126/science.1149213>
- [2] Arenas, A., Díaz-Guilera, A., Kurths, J., Moreno, Y., & Zhou, C. (2008). Synchronization in complex networks. *Physics Reports*, **469**(3), 93–153. <https://doi.org/10.1016/j.physrep.2008.09.002>
- [3] Batthyány, A., & Greyson, B. (2017). Spontaneous remission of dementia before death: Results from a study on paradoxical lucidity. *Psychology of Consciousness: Theory, Research, and Practice*, **8**(1), 1–8. <https://doi.org/10.1037/cns0000259>
- [4] Bola, M., & Sabel, B. A. (2017). Dynamic reorganization of brain functional networks during cognition. *NeuroImage*, **114**, 398–413. <https://doi.org/10.1016/j.neuroimage.2015.03.057>
- [5] Boly, M., Seth, A. K., Wilke, M., Ingmundson, P., Baars, B., Laureys, S., Edelman, D. B., & Tsuchiya, N. (2013). Consciousness in humans and non-human animals: recent advances and future directions. *Frontiers in Psychology*, **4**, 625. <https://doi.org/10.3389/fpsyg.2013.00625>
- [6] Borjigin, J., Lee, U., Liu, T., et al. (2013). Surge of neurophysiological coherence and connectivity in the dying brain. *Proceedings of the National Academy of Sciences*, **110**(35), 14432–14437. <https://doi.org/10.1073/pnas.1308285110>
- [7] Carhart-Harris, R. L., Leech, R., Hellyer, P. J., Shanahan, M., Feilding, A., Tagliazucchi, E., Chialvo, D. R., & Nutt, D. (2014). The entropic brain: a theory of conscious states informed by neuroimaging research with psychedelic drugs. *Frontiers in Human Neuroscience*, **8**, 20. <https://doi.org/10.3389/fnhum.2014.00020>
- [8] Casali, A. G., Gosseries, O., Rosanova, M., et al. (2013). A theoretically based index of consciousness independent of sensory processing and behavior. *Science Translational Medicine*, **5**(198), 198ra105. <https://doi.org/10.1126/scitranslmed.3006294>
- [9] Chennu, S., O'Connor, S., Adapa, R., Menon, D. K., & Bekinschtein, T. A. (2016). Brain

- connectivity dissociates responsiveness from drug exposure during propofol-induced transitions of consciousness. *PLoS Computational Biology*, **12**(1), e1004669. <https://doi.org/10.1371/journal.pcbi.1004669>
- [10] Deco, G., Kringelbach, M. L., Jirsa, V. K., & Ritter, P. (2017). The dynamics of resting fluctuations in the brain: Metastability and its dynamical cortical core. *Scientific Reports*, **7**, 3095. <https://doi.org/10.1038/s41598-017-03073-5>
- [11] Dehaene, S., Kerszberg, M., & Changeux, J.-P. (1998). A neuronal model of a global workspace in effortful cognitive tasks. *PNAS*, **95**(24), 14529–14534. <https://doi.org/10.1073/pnas.95.24.14529>
- [12] Demertzi, A., Tagliazucchi, E., Dehaene, S., et al. (2019). Human consciousness is supported by dynamic complex patterns of brain signal coordination. *Science Advances*, **5**(2), eaat7603. <https://doi.org/10.1126/sciadv.aat7603>
- [13] Fries, P. (2015). Rhythms for cognition: Communication through coherence. *Neuron*, **88**(1), 220–235. <https://doi.org/10.1016/j.neuron.2015.09.034>
- [14] Giacino, J. T., Fins, J. J., Laureys, S., & Schiff, N. D. (2014). Disorders of consciousness after acquired brain injury: the state of the science. *Nature Reviews Neurology*, **10**(2), 99–114. <https://doi.org/10.1038/nrneurol.2013.279>
- [15] Gilmore-Bykovskyi, A., Griffin, J. M., Mueller, K. D., Parnia, S., & Kolanowski, A. (2023). Toward harmonization of strategies for investigating lucidity in AD/ADRD. *Alzheimer's & Dementia*, **19**(1), 343–352. <https://doi.org/10.1002/alz.12709>
- [16] Kelso, J. A. S. (2012). Multistability and metastability: Understanding dynamic coordination in the brain. *Phil. Trans. R. Soc. B*, **367**(1591), 906–918. <https://doi.org/10.1098/rstb.2011.0351>
- [17] Lachaux, J.-P., Rodriguez, E., Martinerie, J., & Varela, F. J. (1999). Measuring phase synchrony in brain signals. *Human Brain Mapping*, **8**(4), 194–208.
- [18] Laureys, S. (2004). Functional neuroimaging in the vegetative state. *NeuroRehabilitation*, **19**(4), 335–341.

- [19] Luppi, A. I., Craig, M. M., Pappas, I., et al. (2019). Consciousness-specific dynamic interactions of brain integration and functional diversity. *Nature Communications*, **10**, 4616. <https://doi.org/10.1038/s41467-019-12658-9>
- [20] Mashour, G. A., Roelfsema, P., Changeux, J.-P., & Dehaene, S. (2020). Conscious processing and the global neuronal workspace hypothesis. *Neuron*, **105**(5), 776–798. <https://doi.org/10.1016/j.neuron.2020.01.026>
- [21] Naccache, L. (2018). Why and how access consciousness can account for phenomenal consciousness. *Phil. Trans. R. Soc. B*, **373**(1755), 20170357. <https://doi.org/10.1098/rstb.2017.0357>
- [22] Nahm, M., Greyson, B., Kelly, E. W., & Haraldsson, E. (2012). Terminal lucidity: A review and a case collection. *Archives of Gerontology and Geriatrics*, **55**(1), 138–142.
- [23] Northoff, G., & Huang, Z. (2017). How do the brain’s time and space mediate consciousness and its different dimensions? *Neuroscience and Biobehavioral Reviews*, **80**, 630–645. <https://doi.org/10.1016/j.neubiorev.2017.07.013>
- [24] Restrepo, J. G., Ott, E., & Hunt, B. R. (2005). Onset of synchronization in large networks of coupled oscillators. *Physical Review E*, **71**(3), 036151.
- [25] Sarasso, S., Boly, M., Napolitani, M., et al. (2015). Consciousness and complexity during unresponsiveness induced by propofol, xenon, and ketamine. *Current Biology*, **25**(23), 3099–3105. <https://doi.org/10.1016/j.cub.2015.10.014>
- [26] Schartner, M., Seth, A., Noirhomme, Q., et al. (2015). Complexity of multi-dimensional spontaneous EEG decreases during propofol induced general anaesthesia. *PLoS ONE*, **10**(8), e0133532. <https://doi.org/10.1371/journal.pone.0133532>
- [27] Sitt, J. D., King, J.-R., El Karoui, I., et al. (2014). Large scale screening of neural signatures of consciousness in patients in a vegetative or minimally conscious state. *Brain*, **137**(8), 2258–2270.
- [28] Stam, C. J. (2005). Nonlinear dynamical analysis of EEG and MEG: Review of an emerging field. *Clinical Neurophysiology*, **116**(10), 2266–2301.

- [29] Szalkai, B., Kerepesi, C., Varga, B., & Grolmusz, V. (2015). The Budapest Reference Connectome Server v2.0. *Neuroscience Letters*, **595**, 60–62. <https://doi.org/10.1016/j.neulet.2015.03.071>
- [30] Szalkai, B., Kerepesi, C., Varga, B., & Grolmusz, V. (2017). Parameterizable consensus connectomes from the Human Connectome Project: The Budapest Reference Connectome Server v3.0. *Cognitive Neurodynamics*, **11**(1), 113–116. <https://doi.org/10.1007/s11571-016-9407-z>
- [31] Tagliazucchi, E. (2017). The signatures of conscious access and its phenomenology are consistent with large-scale brain communication at criticality. *Consciousness and Cognition*, **55**, 136–147. <https://doi.org/10.1016/j.concog.2017.08.008>
- [32] Tononi, G. (2004). An information integration theory of consciousness. *BMC Neuroscience*, **5**, 42.
- [33] Xu, G., Mihaylova, T., Li, D., et al. (2023). Surge of neurophysiological coupling and connectivity of gamma oscillations in the dying human brain. *Proceedings of the National Academy of Sciences*, **120**(19), e2216268120. <https://doi.org/10.1073/pnas.2216268120>

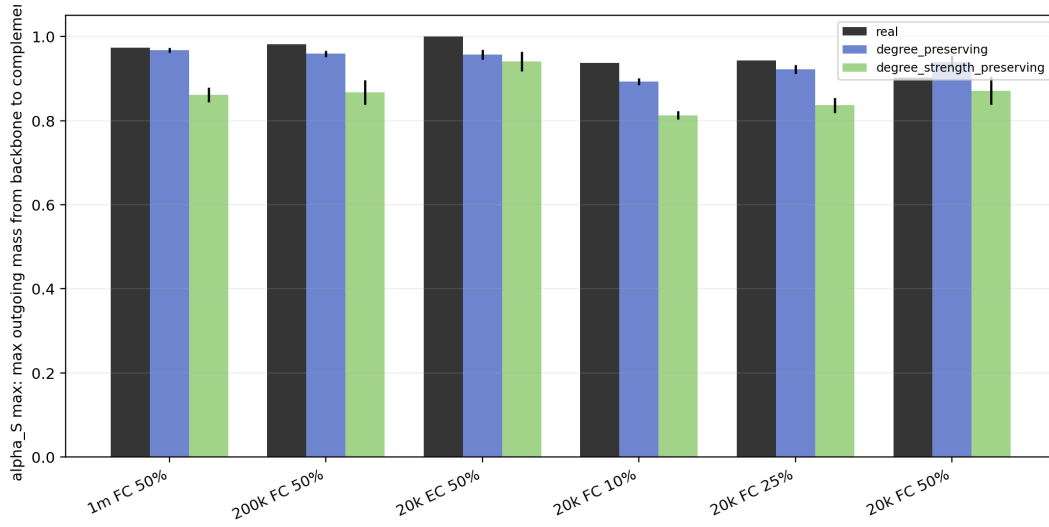


Figure 6: Structural external-mass diagnostic for the HCP-derived connectome stress test. Lower α_S would support the conservative sufficient-condition intuition that the residual backbone is weakly perturbed by the complement. Empirical top-strength backbones did not generally show lower α_S than the stricter null graphs.

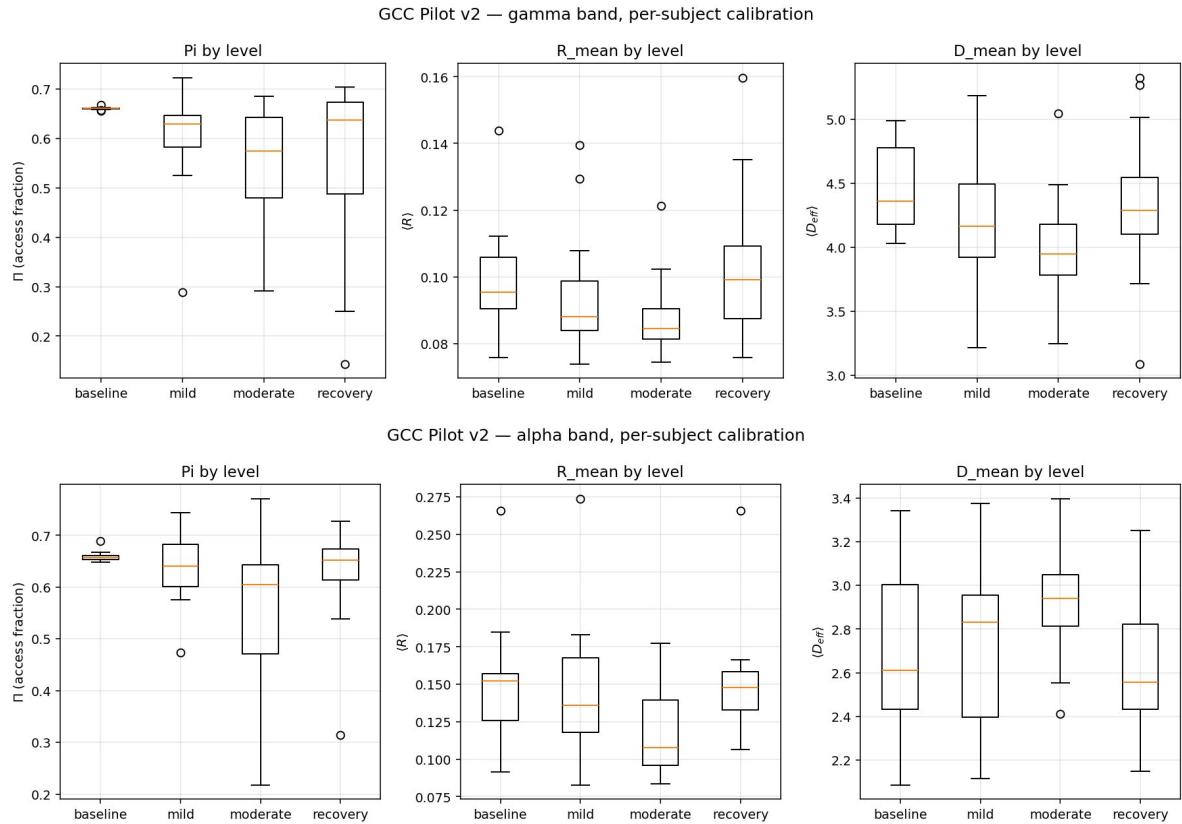


Figure 7: Access indicator Π , order parameter $\langle R \rangle$, and effective dimensionality per sedation level ($N = 20$, per-subject calibration). Top: gamma band. Bottom: alpha band.